Venkata Lakshmi Parimala Pasupuleti

Backend AI Engineer · Multi-Agent Orchestration · LangGraph · LLM Evaluation · FastAPI · Python · GCP · AWS

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PROFESSIONAL SUMMARY

Backend AI engineer with 7+ years production experience building the systems behind financial services at scale — now shipping **multi-agent orchestration, LLM evaluation harnesses, and model routing** at Intuit. Built AgentProbe, an open-source **agent runtime** with a smart LLM Router (dynamic model routing across 5 providers), 3 LangGraph agents, RAG evaluation, and **LLM-as-judge CI harness** — live at github.com/venkatalakshmiparimala/AgentProbe. Built LeaseI: an 8-agent LangGraph platform (triage → retrieval → reasoning → validation → HITL escalation) with **prompt caching, context engineering, hybrid-search RAG**, and **eval harness achieving 93.6% precision across 1,240 conversations**. At Entrata, built **async processing with RabbitMQ** (pub/sub, idempotency, retry logic) across thousands of tenant records at **99.5% uptime**; optimized **200+ SQL queries for 30% performance gains**; Docker microservices running at production scale on AWS. Daily Claude Code user. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Agents: Multi-Agent Orchestration, LangGraph, LangChain, LLM Router, Model Routing, Prompt Caching, Context Engineering, RAG Pipelines, Tool Calling, LLM-as-Judge, Eval Harnesses, Conversation Replay, HITL Escalation, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, HuggingFace

Backend & APIs: Python (9+ yrs), FastAPI, Flask, Java, Spring Boot, TypeScript, Node.js, REST APIs, Async Processing, Pub/Sub (RabbitMQ), Idempotency, Retries, Caching, Celery, GraphQL

Cloud & Infrastructure: GCP/Vertex AI, AWS (S3, EC2, RDS, Lambda, SQS, Glue, API Gateway, IAM), Docker, Kubernetes, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, Terraform

Data & Auth: PostgreSQL, MySQL, Snowflake, Redis, MongoDB, SQLite, ChromaDB, HIPAA/PHI RBAC, Fine-Grained Authorization, SQL Optimization

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Backend AI · Multi-Agent Systems · Financial Analytics

- Build and operate **multi-agent orchestration** for financial analytics AI: Python/FastAPI services integrating Anthropic Claude with **model routing, prompt caching**, and fine-grained authorization; add guardrails (input validation, sensitive-data handling, fallbacks) and ship via GitHub Actions CI/CD to AWS with Prometheus/Grafana for production observability.
- Build and run **eval harnesses and LLM-as-judge pipelines** (golden datasets + regression suites) to measure every model and prompt change before it ships — catching hallucination and quality drift; build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake from feature engineering through prediction APIs.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Backend Systems · Async Processing · Data Pipelines

- Built **RabbitMQ async processing system** for rent updates across thousands of tenant records — pub/sub architecture with **idempotency, retry logic, and dead-letter handling** to prevent double processing; **99.5% uptime** across a 100K+ MAU multi-tenant SaaS platform.
- Shipped 20+ backend features end-to-end (Python, PHP, Java, JavaScript, PostgreSQL); **optimized 200+ SQL queries (30% performance improvement)** across 1M+ records; Docker microservices redesign cutting execution time 50% and deployments 30% faster; AWS S3 migration with **Libsodium encryption and RBAC** for resident PHI — **100% compliance, zero incidents**; reviewed 300+ PRs, mentored junior engineers; delivered across **120+ Agile sprints**.

PROJECTS

AgentProbe — Open-Source Agent Runtime & LLM Eval Toolkit

Multi-Agent · LLM Router · Evals

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- **Agent runtime** with dynamic **model routing** across 5 LLM providers, 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix), **tool calling**, and RAG pipeline evaluation; **LLM-as-judge eval harness** with CI golden-dataset regression testing; 15-endpoint FastAPI backend; 5-metric scoring (factual accuracy, hallucination rate, retrieval quality, completeness, compliance). Every model and prompt change measured before ship.

LeaseI — 8-Agent LangGraph Conversational AI Platform

Orchestration · RAG · HITL

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG · Prompt Caching · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Multi-agent orchestration** with real conditional edges (triage → retrieval → reasoning → validation → HITL escalation); **prompt caching, context engineering**, hybrid-search RAG (TF-IDF + semantic reranking); **LLM-as-judge eval harness: 93.6% precision, 87.9% F1** across 1,240 conversations; fine-grained authorization enforcing which agent can take which action — LLM reasons, deterministic logic decides every regulated call.

AI on FHIR — NLP-to-API Conversational Backend

Backend AI · Auth · Multilingual

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs — github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates EN/ES natural-language queries into FHIR API calls using LLMs + BERT + embeddings; **role-based access control for HIPAA-style fine-grained authorization**; Docker containerized; cut manual clinical lookup time ~60%.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, LLM evaluation, distributed systems, statistical modeling, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017